

Faculty of Electrical Engineering

WARSAW UNIVERSITY OF TECHNOLOGY

Lighting Design of Public Spaces

PROJECT

TITLE

Outdoor car Parking

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Introduction

This project is the lighting design of outdoor parking space. My dimensions are 80mx45m and I chose it to be of medium traffic type. We can assume it is a department store(big store, mega market). My parking is symmetrical regarding both axes. So we enter the parking from the road side and from the other end we enter the building of interest. Below table 1 is shown where lighting requirements are written down. Medium traffic means we try to get to 10lx and uniformity of 0.25.

Parking Area Category	Maintained Illuminance (E_m) [lx]	Uniformity (U_o) [-]	Glare Limit (RGL) [-]	Colour Rendering Index (R_a) [-]
Light Traffic (Residential, small shops, bicycle parks)	5	0.25	55	20
Medium Traffic (Office buildings, department stores)	10	0.25	50	20
Heavy Traffic (Large shopping centers, major hubs)	20	0.25	50	20

Table 1 - requirements

Work and simulation

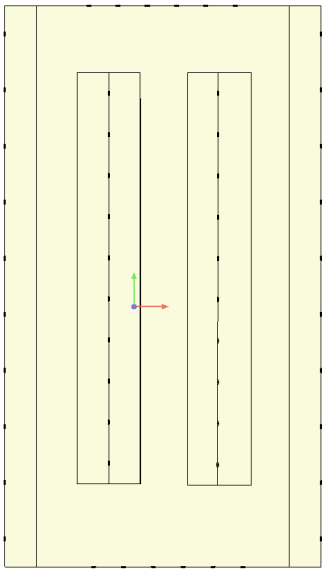


Figure 1 - schematic of parking

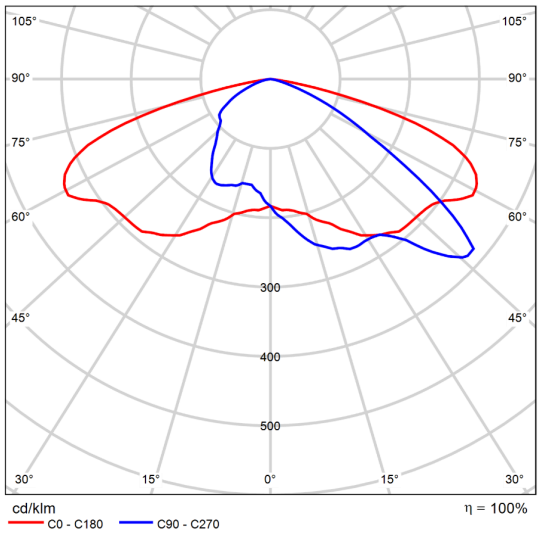
Luminiare used:
Trilux LIQ 50N-AB7L-1700-7302 1 06
Luminous flux - 1700lx
Power 13.5

This exact luminaire had a customizing feature, meaning I myself chose everything related to it, maybe thats why it is now hard to find its data sheet.

Photo



LID



CCT - 3000K
CRI - 70

Here below figure 2 we can observe the luminaires locations precisely. There are 52 luminaires of this one type.

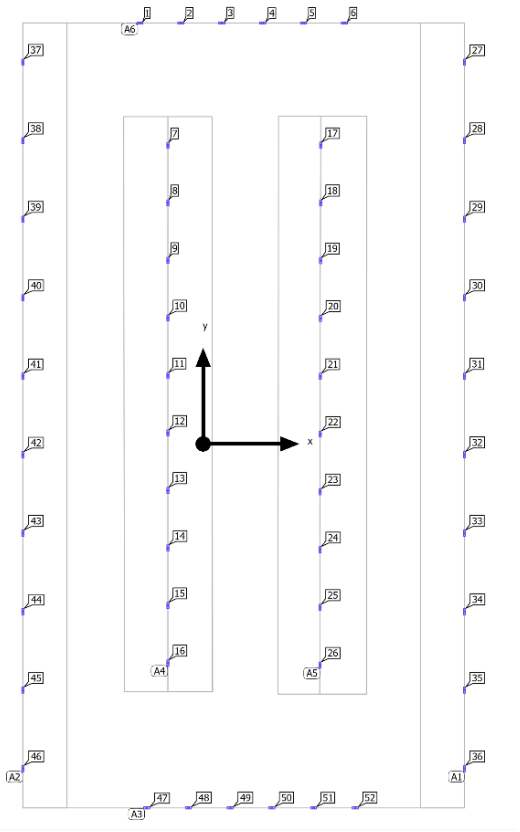


Figure 2

Here is the calculation surfaces showing average luminance values.

Calculation surfaces

Properties	$\bar{E}$	E_{min}	E_{max}	U_o (g_1)
Calculation surface 1 Perpendicular illuminance Height: 0.000 m	13.4 lx	3.64 lx	25.9 lx	0.27
Calculation surface 1 Cylindrical illuminance Height: 1.200 m	7.06 lx	1.85 lx	12.2 lx	0.26

Next up let me show the Glare rating which should be under 50, which in our case is 43.

Calculation surface 1 (R_G)

Strongest glare at	255°
max	43
Target	≤ 50
Viewing sector	0° - 360°
Step width	15°
Angle of inclination	-2°
Height	1.200 m
Index	CG1
Method	Simplified calculation according to EN 12464

Summary

This projects aim was to create a draft lighting design for outdoor parking space, while trying to get to as much detail as needed for the course. And design which satisfies all the requirements has been developed and simulated in Dialux evo. Which proved that theoretically the project is good.